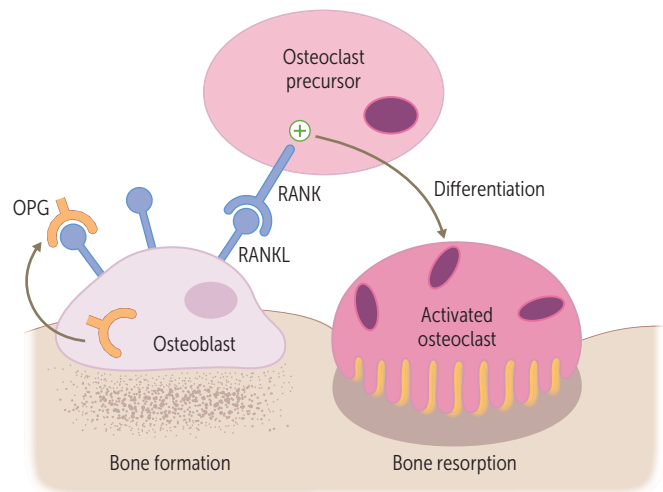


Cell biology of bone

Osteoblast	Builds bone by secreting collagen and catalyzing mineralization in alkaline environment via ALP. Differentiates from mesenchymal stem cells in periosteum. Osteoblastic activity measured by bone ALP, osteocalcin, propeptides of type I procollagen.
Osteoclast	Dissolves (“crushes”) bone by secreting H^+ and collagenases. Differentiates from a fusion of monocyte/macrophage lineage precursors. RANK receptors on osteoclasts are stimulated by RANKL (RANK ligand, expressed on osteoblasts). OPG (osteoprotegerin, a RANKL decoy receptor) binds RANKL to prevent RANK-RANKL interaction → ↓ osteoclast activity.
Parathyroid hormone	At low, intermittent levels, exerts anabolic effects (building bone) on osteoblasts and osteoclasts (indirect). Chronically ↑ PTH levels (1° hyperparathyroidism) cause catabolic effects (osteitis fibrosa cystica).
Estrogen	Inhibits apoptosis in bone-forming osteoblasts and induces apoptosis in bone-resorbing osteoclasts. Causes closure of epiphyseal plate during puberty. Estrogen deficiency (surgical or postmenopausal) → ↑ cycles of remodeling and bone resorption → ↑ risk of osteoporosis.

**► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—PATHOLOGY****Overuse injuries of the elbow**

Medial (golfer's) elbow tendinopathy	Repetitive wrist flexion or idiopathic → pain near medial epicondyle.
Lateral (tennis) elbow tendinopathy	Repetitive wrist extension (backhand shots) or idiopathic → pain near lateral epicondyle.

Clavicle fractures

Common in children and as birth trauma. Usually caused by a fall on outstretched hand or by direct trauma to shoulder. Weakest point at the junction of middle and lateral thirds **A**; fractures at the middle third segment are most common. Presents as shoulder drop, shortened clavicle (lateral fragment is depressed due to arm weight and medially rotated by arm adductors [eg, pectoralis major]).

**Wrist and hand injuries****Guyon canal syndrome**

Compression of ulnar nerve at wrist. Classically seen in cyclists due to pressure from handlebars.

May also be seen with fracture/dislocation of the hook of hamate.

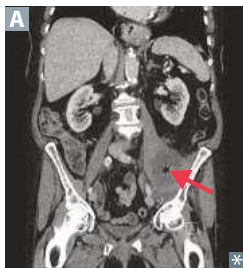
Carpal tunnel syndrome

Entrapment of median nerve in carpal tunnel (between transverse carpal ligament and carpal bones) → nerve compression → paresthesia, pain, and numbness in distribution of median nerve. Thenar eminence atrophies but sensation spared, because palmar cutaneous branch enters hand external to carpal tunnel.

Suggested by ⊕ Tinel sign (percussion of wrist causes tingling) and Phalen maneuver (90° flexion of wrist causes tingling). Associated with pregnancy (due to edema), rheumatoid arthritis, hypothyroidism, diabetes, acromegaly, dialysis-related amyloidosis; may be associated with repetitive use.

Metacarpal neck fracture

Most commonly the 5th metacarpal **A** ("boxer's fracture"). Typically occurs in young males due to direct trauma (eg, punching a wall). Presents with pain, swelling, and tenderness over the affected metacarpal; reduced grip strength; possible deformity.

Psoas abscess

Collection of pus in iliopsoas compartment. May spread from blood (hematogenous) or from adjacent structures (eg, vertebral osteomyelitis, tuberculous spondylitis [also called Pott disease], pyelonephritis). Associated with Crohn disease, diabetes, and immunocompromised states.

Staphylococcus aureus most commonly isolated, but may also occur 2° to tuberculosis.

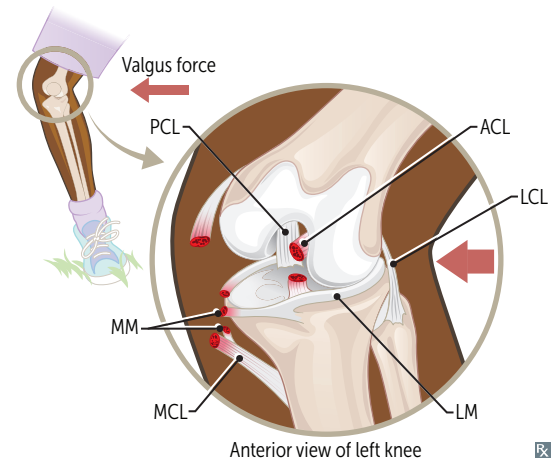
Findings: flank pain, fever, inguinal mass, ⊕ psoas sign (hip extension exacerbates lower abdominal pain).

Labs: leukocytosis. Imaging (CT/MRI) will show focal hypodense lesion within the muscle plane (red arrow in **A**).

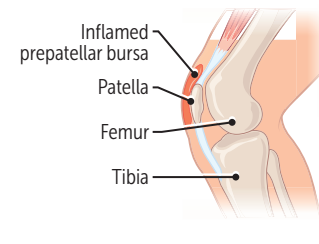
Treatment: abscess drainage, antibiotics.

Common knee conditions**“Unhappy triad”**

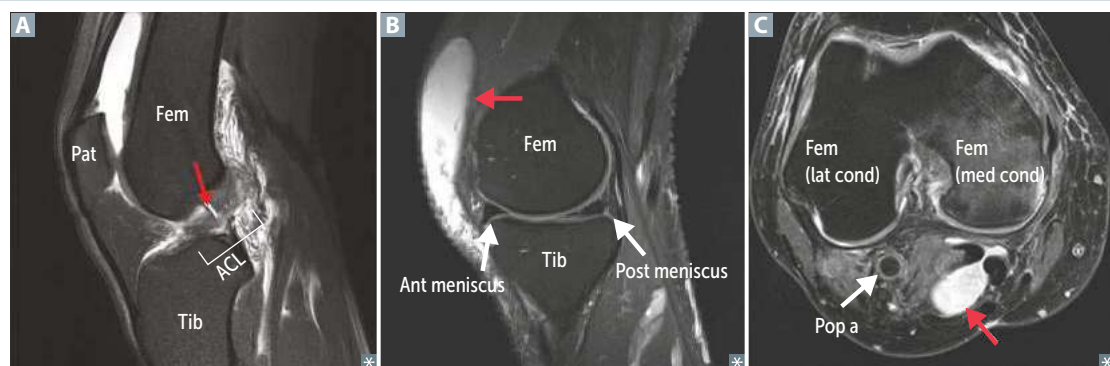
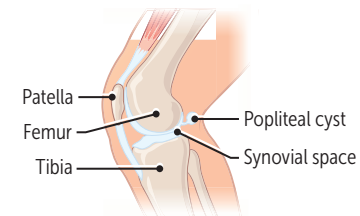
Common injury in contact sports due to lateral force impacting the knee when foot is planted on the ground. Consists of damage to the ACL **A**, MCL, and medial meniscus (attached to MCL). However, lateral meniscus involvement is more common than medial meniscus involvement in conjunction with ACL and MCL injury. Presents with acute pain and signs of joint instability.

**Prepatellar bursitis**

Inflammation of the prepatellar bursa in front of the kneecap (red arrow in **B**). Can be caused by repeated trauma or pressure from excessive kneeling (also called “housemaid’s knee”).

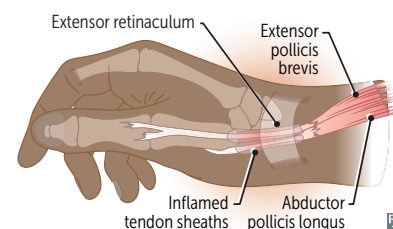
**Popliteal cyst**

Also called Baker cyst. Popliteal fluid collection (red arrow in **C**) in gastrocnemius-semimembranosus bursa commonly communicating with synovial space and related to chronic joint disease (eg, osteoarthritis, rheumatoid arthritis).



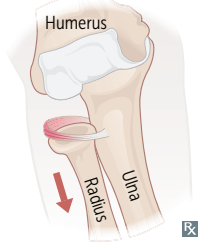
Common musculoskeletal conditions

Costochondritis	Inflammation of costochondral or costosternal junctions. Presents with focal tenderness to palpation and sharp, positional chest pain that may worsen with deep inspiration. More common in younger female patients. May mimic cardiac (eg, MI) or pulmonary (eg, pulmonary embolism) diseases.
De Quervain tenosynovitis	Noninflammatory thickening of abductor pollicis longus and extensor pollicis brevis tendons → pain or tenderness at radial styloid. ⊕ Finkelstein test (pain at radial styloid with active or passive stretch of thumb tendons). ↑ risk in new mothers (lifting baby), golfers, racquet sport players, “thumb” texters.
Dupuytren contracture	Caused by fibroblastic proliferation and thickening of superficial palmar fascia. Typically involves the fascia at the base of the ring and little fingers. Unknown etiology; most frequently seen in males > 50 years old of Northern European descent.
Ganglion cyst	Mucin-filled swelling overlying joint or tendon sheath, most commonly at dorsal side of wrist. Transilluminates with shining light (tumors lack transillumination). Usually resolves spontaneously.
Iliotibial band syndrome	Overuse injury of lateral knee that occurs primarily in runners. Pain develops 2° to friction of iliotibial band against lateral femoral epicondyle.
Limb compartment syndrome	↑ pressure within fascial compartment of a limb → venous outflow obstruction and arteriolar collapse → anoxia, necrosis, rhabdomyolysis → acute tubular necrosis. Causes include significant long bone fractures (eg, tibia), reperfusion injury, animal venoms. Presents with severe pain and tense, swollen compartments with passive stretch of muscles in the affected compartment. Increased serum creatine kinase and motor deficits are late signs of irreversible muscle and nerve damage. 5 P's: pain, pallor, paresthesia, pulselessness, paralysis.
Medial tibial stress syndrome	Also called shin splints. Common cause of shin pain and diffuse tenderness in runners and military recruits. Caused by bone resorption that outpaces bone formation in tibial cortex.
Plantar fasciitis	Inflammation of plantar aponeurosis characterized by heel pain (worse with first steps in the morning or after period of inactivity) and tenderness. Associated with obesity, prolonged standing or jumping (eg, dancers, runners), and flat feet. Heel spurs often coexist.
Temporomandibular disorders	Group of disorders that involve the temporomandibular joint (TMJ) and muscles of mastication. Multifactorial etiology; associated with TMJ trauma, poor head and neck posture, abnormal trigeminal nerve pain processing, psychological factors. Present with dull, constant unilateral facial pain that worsens with jaw movement, otalgia, headache, TMJ dysfunction (eg, limited range of motion).



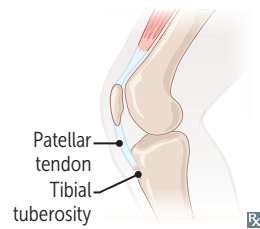
Childhood musculoskeletal conditions

Radial head subluxation



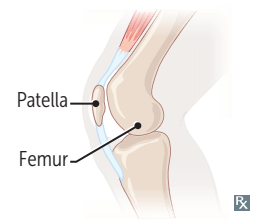
Also called nursemaid's elbow. Common elbow injury in children < 5 years. Caused by a sudden pull on the arm → immature annular ligament slips over head of radius. Injured arm held in slightly flexed and pronated position.

Osgood-Schlatter disease



Also called traction apophysitis. Overuse injury caused by repetitive strain and chronic avulsion of the secondary ossification center of the tibial tuberosity. Occurs in adolescents after growth spurt. Common in athletes who run and jump. Presents with progressive anterior knee pain.

Patellofemoral syndrome



Overuse injury that commonly presents in young, female athletes as anterior knee pain. Exacerbated by prolonged sitting or weight-bearing on a flexed knee.

Developmental dysplasia of the hip

Abnormal acetabulum development in newborns. Risk factor is breech presentation. Results in hip instability/dislocation. Commonly tested with Ortolani and Barlow maneuvers (manipulation of newborn hip reveals a “clunk”). Confirmed via ultrasound (x-ray not used until ~4–6 months because cartilage is not ossified).

Legg-Calvé-Perthes disease

Idiopathic avascular necrosis of femoral head. Commonly presents between 5–7 years with insidious onset of hip pain that may cause child to limp. More common in males (4:1 ratio). Initial x-ray often normal.

Slipped capital femoral epiphysis



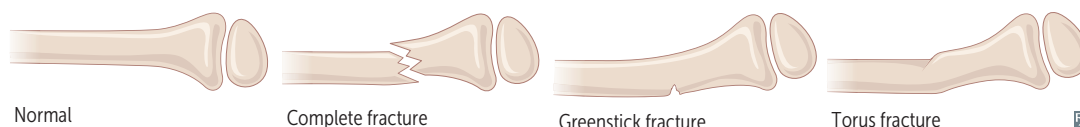
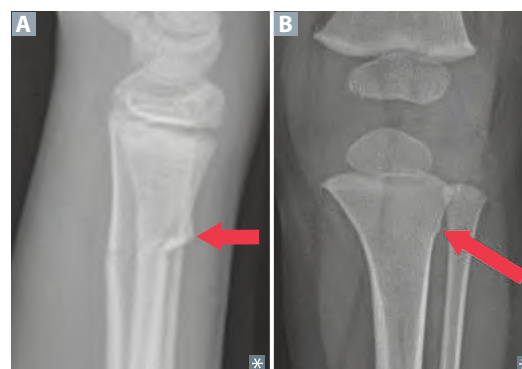
Classically presents in an obese young adolescent with hip/knee pain and altered gait. Increased axial force on femoral head → epiphysis displaces relative to femoral neck (like a scoop of ice cream slipping off a cone). Diagnosed via x-ray **A**.

Common pediatric fractures**Greenstick fracture**

Incomplete fracture extending partway through width of bone **A** following bending stress; bone fails on tension side; compression side intact (compare to torus fracture). Bone is bent like a **green twig**.

Torus (buckle) fracture

Axial force applied to immature bone → cortex buckles on compression (concave) side and fractures **B**. Tension (convex) side **remains solid** (intact).

**Achondroplasia**

Failure of longitudinal bone growth (endochondral ossification) → short limbs. Membranous ossification is not affected → large head relative to limbs. Constitutive activation of fibroblast growth factor receptor (FGFR3) actually inhibits chondrocyte proliferation. > 85% of mutations occur sporadically; autosomal dominant with full penetrance (homozygosity is lethal). Associated with ↑ paternal age. Most common cause of short-limbed dwarfism.

Osteoporosis

Trabecular (spongy) and cortical bone lose mass despite normal bone mineralization and lab values (serum Ca^{2+} and PO_4^{3-}).

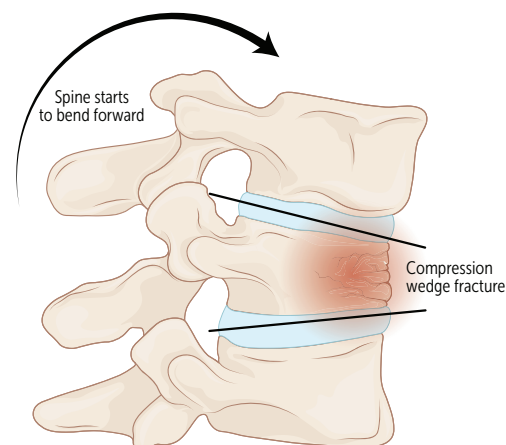
Most commonly due to ↑ bone resorption (↑ osteoclast number and activity) related to ↓ estrogen levels, old age, and cigarette smoking. Can be 2° to drugs (eg, steroids, alcohol, anticonvulsants, anticoagulants, thyroid replacement therapy) or other conditions (eg, hyperparathyroidism, hyperthyroidism, multiple myeloma, malabsorption syndromes, anorexia), low BMI (or weight), and prolonged microgravity exposure (eg, space travel).

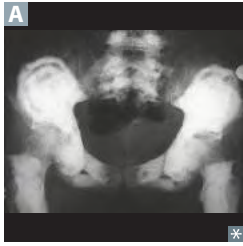
Diagnosed by bone mineral density measurement by DEXA (dual-energy x-ray absorptiometry) at the lumbar spine, total hip, and femoral neck, with a T-score of ≤ -2.5 or by a fragility fracture (eg, fall from standing height, minimal trauma) at hip or vertebra. One-time screening recommended in females ≥ 65 years old.

Prophylaxis: regular weight-bearing exercise and adequate Ca^{2+} and vitamin D intake throughout adulthood.

Treatment: bisphosphonates, teriparatide/abaloparatide, SERMs, denosumab (monoclonal antibody against RANKL), romosozumab (sclerostin inhibitor).

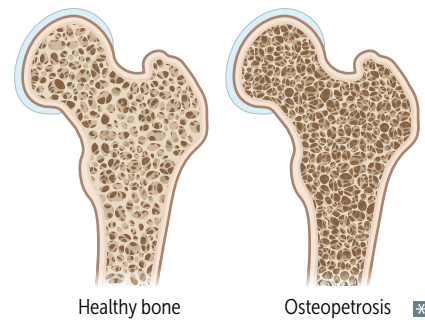
Can lead to **vertebral compression fractures** **A**—acute back pain, loss of height, kyphosis. Also can present with fractures of femoral neck, distal radius (Colles fracture).



Osteopetrosis

Failure of normal bone resorption due to defective osteoclasts → thickened, dense bones that are prone to fracture. Mutations (eg, carbonic anhydrase II) impair ability of osteoclast to generate acidic environment necessary for bone resorption. Overgrowth of cortical bone fills marrow space → pancytopenia, extramedullary hematopoiesis. Can result in cranial nerve impingement and palsies due to narrowed foramina.

X-rays show diffuse symmetric sclerosis (bone-in-bone, “stone bone” **A**). Bone marrow transplant is potentially curative as osteoclasts are derived from monocytes.

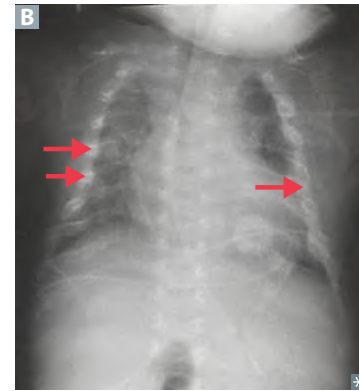
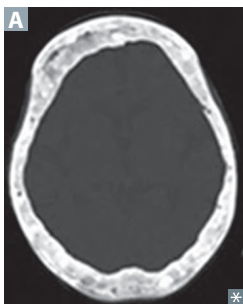
**Osteomalacia/rickets**

Defective mineralization of osteoid (osteomalacia) or cartilaginous growth plates (rickets, only in children). Most commonly due to vitamin D deficiency.

X-rays show osteopenia and pseudofractures in osteomalacia, epiphyseal widening and metaphyseal cupping/fraying in rickets. Children with rickets have pathologic bow legs (genu varum **A**), beadlike costochondral junctions (rachitic rosary **B**), craniotabes (soft skull).

↓ vitamin D → ↓ serum Ca^{2+} → ↑ PTH secretion → ↓ serum PO_4^{3-} .

Hyperactivity of osteoblasts → ↑ ALP.

**Osteitis deformans**

Also called Paget disease of bone. Common, localized disorder of bone remodeling caused by ↑ osteoclastic activity followed by ↑ osteoblastic activity that forms poor-quality bone. Serum Ca^{2+} , phosphorus, and PTH levels are normal. ↑ ALP. Mosaic pattern of woven and lamellar bone (osteocytes within lacunae in chaotic juxtapositions); long bone chalk-stick fractures. ↑ blood flow from ↑ arteriovenous shunts may cause high-output heart failure. ↑ risk of osteosarcoma.

Hat size can be increased due to skull thickening **A**; hearing loss is common due to skull deformity.

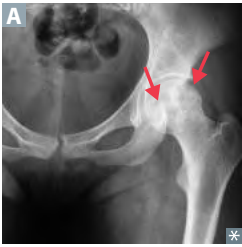
Stages of Paget disease:

- Early destructive (lytic): osteoclasts
- Intermediate (mixed): osteoclasts + osteoblasts
- Late (sclerotic/blastic): osteoblasts

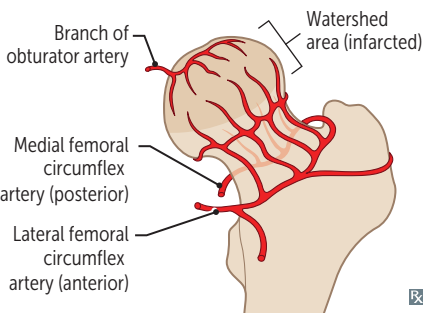
May enter quiescent phase.

Treatment: bisphosphonates.

Avascular necrosis of bone



Infarction of bone and marrow, usually very painful. Most common site is femoral head (watershed area) **A** (due to insufficiency of medial circumflex femoral artery). Causes include glucocorticoids, chronic alcohol overuse, sickle cell disease, trauma, SLE, “the Bends” (caisson/decompression disease), Legg-Calvé-Perthes disease (idiopathic), Gaucher disease, slipped capital femoral epiphysis—**CASTS Bend LEGS**.



Lab values in bone disorders

DISORDER	SERUM Ca^{2+}	PO_4^{3-}	ALP	PTH	COMMENTS
Osteoporosis	—	—	—	—	↓ bone mass; if concurrent ↓ vitamin D → ↑ PTH with normal Ca^{2+}
Osteopetrosis	—/↓	—	—	—	Dense, brittle bones. Ca^{2+} ↓ in severe, malignant disease
Paget disease of bone	—	—	↑	—	Abnormal “mosaic” bone architecture
Osteitis fibrosa cystica Primary hyperparathyroidism	↑	↓	↑	↑	“Brown tumors” due to fibrous replacement of bone, subperiosteal thinning Idiopathic or parathyroid hyperplasia, adenoma, carcinoma
Secondary hyperparathyroidism	↓	↑	↑	↑	Often as compensation for CKD (↓ PO_4^{3-} excretion and production of activated vitamin D)
Osteomalacia/rickets	↓	↓	↑	↑	Soft bones; vitamin D deficiency also causes 2° hyperparathyroidism
Hypervitaminosis D	↑	↑	—	↓	Caused by oversupplementation or granulomatous disease (eg, sarcoidosis)

↑ ↓ = 1° change.

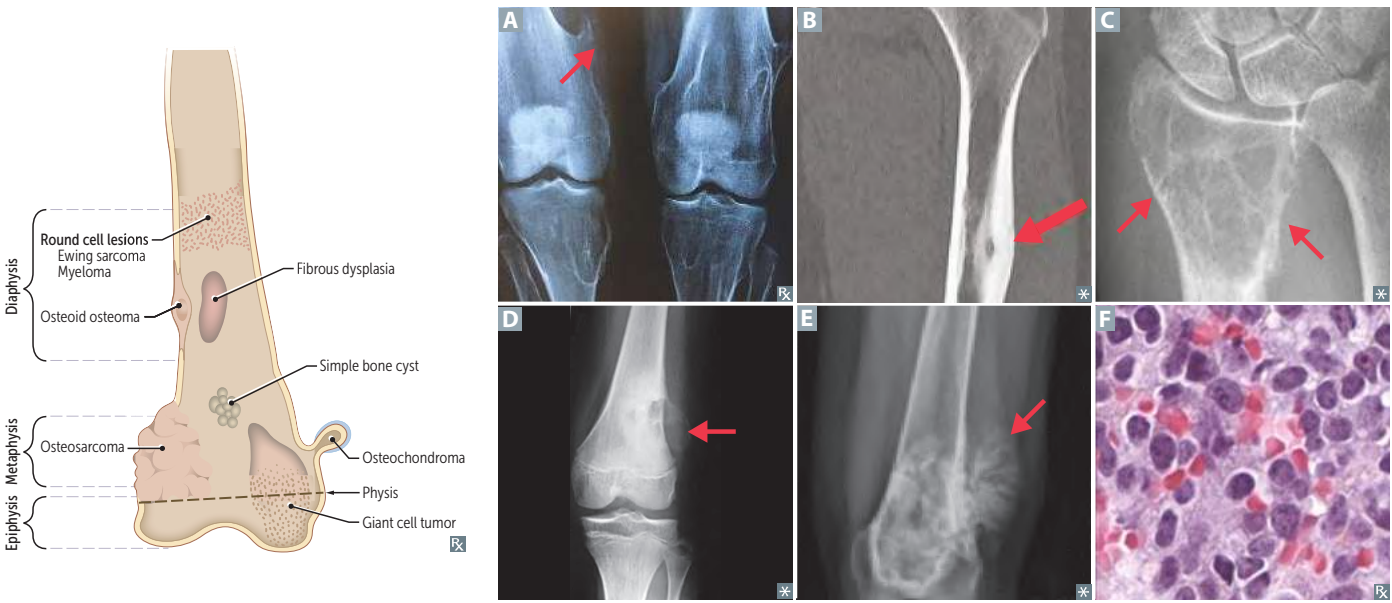
Primary bone tumors

Metastatic disease is more common than 1° bone tumors. **B**enign **b**one tumors end with -oma, those that start with a **c** and **o** are more **c**ommon in **b**oys. Malignant tumors usually have the ending “-sarcoma”.

TUMOR TYPE	EPIDEMIOLOGY	LOCATION	CHARACTERISTICS
Benign tumors			
Osteochondroma (exostosis)	Most common benign bone tumor. Males < 25 years old.	Metaphysis of long bones (most common around knee—distal femur).	Lateral bony projection of growth plate (continuous with marrow space) covered by cartilaginous cap A ; points away from joint. EXT1 or EXT2 gene mutation—hereditary multiple exostoses . Rarely transforms to chondrosarcoma.
Osteoma	Middle age.	Surface of facial bones.	Associated with Gardner syndrome.
Osteoid osteoma	Adults < 25 years old. Males > females.	Cortex of long bones.	Classically presents as bone pain (worse at night) caused by prostaglandins; thus relieved by NSAIDs (vs osteoblastoma). Bony mass (< 1.5 cm) with radiolucent osteoid core B .
Osteoblastoma	Males > females.	Vertebrae.	Similar histology to osteoid osteoma Larger size (> 2 cm); pain unresponsive to NSAIDs. X-ray similar to aneurysmal bone cyst.
Giant cell tumor	20–40 years old. Females > males.	Epiphysis of long bones after skeletal maturation (often in knee region), radiographic epicenter is metaphysis.	Locally aggressive benign tumor neoplastic mononuclear cells that express RANKL and reactive multinucleated giant (osteoclastlike) cells; “osteoclastoma”. “Soap bubble” appearance on x-ray C .
Chondroblastoma	Adolescents. Males > females.	Epiphysis of long bones before skeletal maturation (often in knee region).	May complain of joint pain. Cross physis on x-ray.

Primary bone tumors (continued)

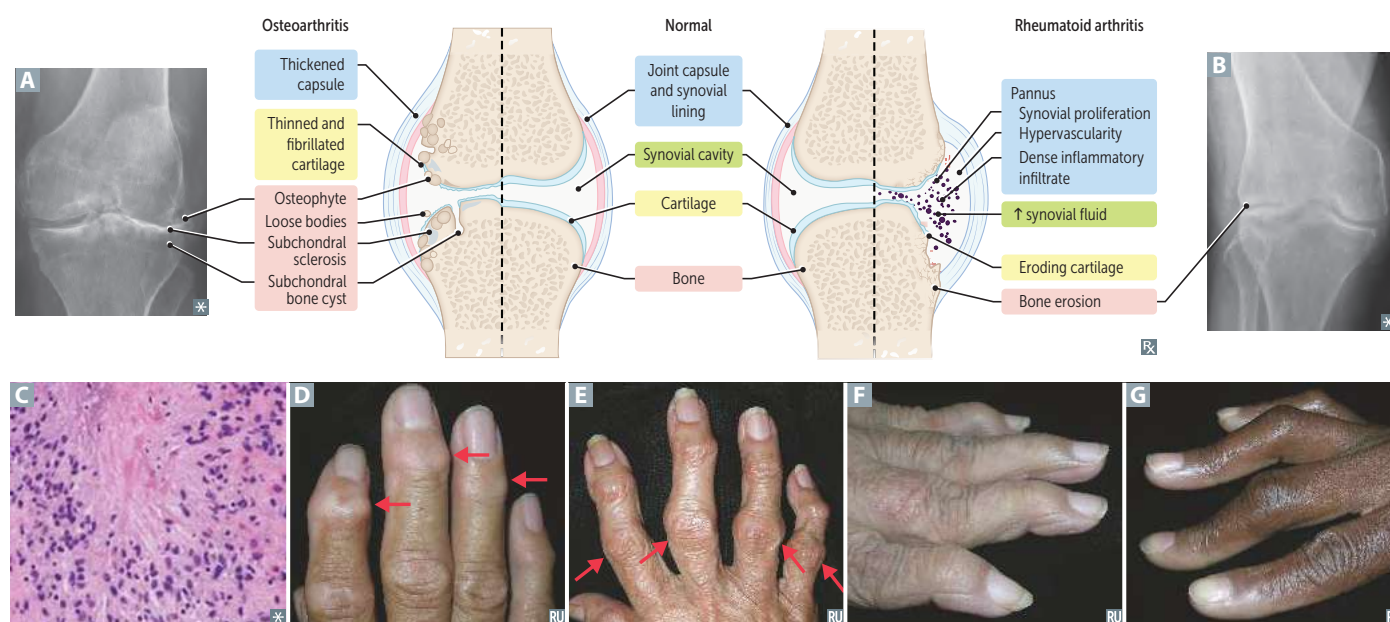
TUMOR TYPE	EPIDEMIOLOGY	LOCATION	CHARACTERISTICS
Malignant tumors			
Osteosarcoma (osteogenic sarcoma)	Accounts for 20% of 1° bone cancers. Peak incidence of 1° tumor in males < 20 years. Less common in older adults; usually 2° to predisposing factors, such as Paget disease of bone, bone infarcts, radiation, familial retinoblastoma, Li-Fraumeni syndrome.	Metaphysis of long bones (often in knee region).	Pleomorphic osteoid-producing cells (malignant osteoblasts). Presents as painful enlarging mass or pathologic fractures. Codman triangle D (from elevation of periosteum) or sunburst pattern on x-ray E (think of an osteocod [bone fish] swimming in the sun). Aggressive. 1° usually responsive to treatment (surgery, chemotherapy), poor prognosis for 2°.
Chondrosarcoma	Most common in adults > 50 years old.	Medulla of pelvis, proximal femur and humerus.	Tumor of malignant chondrocytes. Lytic (> 50%) cases with intralesional calcifications, endosteal erosion, cortex breach.
Ewing sarcoma	Most common in White patients, generally males < 15 years old.	Diaphysis of long bones (especially femur), pelvic flat bones.	Anaplastic small blue cells of neuroectodermal (mesenchymal) origin (resemble lymphocytes) F . Differentiate from conditions with similar morphology (eg, lymphoma, chronic osteomyelitis) by testing for t(11;22) (fusion protein EWS-FLI1). “Onion skin” periosteal reaction. Aggressive with early metastases, but responsive to chemotherapy. 11 + 22 = 33 (Patrick Ewing ’s jersey number).



Osteoarthritis vs rheumatoid arthritis

	Osteoarthritis A	Rheumatoid arthritis B
PATHOGENESIS	Chronic mechanical stress destroys articular cartilage → inflammation with inadequate repair (mediated by chondrocytes).	Autoimmune—inflammation C induces formation of proliferative granulation tissue, eroding articular cartilage and bone.
PREDISPOSING FACTORS	Age, female, obesity, joint trauma.	Female, HLA-DR4 (4-walled “ rheum ”), HLA-DRB1, smoking. ⊕ rheumatoid factor (IgM antibody that targets IgG Fc; in 80%), anti-cyclic citrullinated peptide antibody (more specific).
PRESENTATION	Pain in weight-bearing joints after use, improving with rest. Asymmetric involvement. No systemic symptoms.	Pain, swelling, and morning stiffness lasting > 1 hour, improving with use. Symmetric involvement. Systemic symptoms (fever, fatigue, weight loss). Extraarticular manifestations.*
JOINT FINDINGS	Osteophytes (bone spurs), joint space narrowing (asymmetric), subchondral sclerosis and cysts, loose bodies. Synovial fluid noninflammatory (WBC < 2000/mm ³). Development of Heberden nodes D (at DIP) and Bouchard nodes E (at PIP), and 1st CMC.	Erosions, juxta-articular osteopenia, soft tissue swelling, subchondral cysts, joint space narrowing (symmetric). Deformities: cervical subluxation, ulnar finger deviation, swan neck F , boutonniere G . Involves MCP, PIP, wrist.
TREATMENT	Activity modification, NSAIDs, intra-articular glucocorticoids (use for short-term relief in symptomatic patients; long-term therapy associated with many adverse effects).	NSAIDs, glucocorticoids, disease-modifying agents (eg, methotrexate, sulfasalazine), biologic agents (eg, TNF- α inhibitors).

*Cervical subluxation, rheumatoid nodules (fibrinoid necrosis with palisading histiocytes) in subcutaneous tissue and lung (+ pneumoconiosis = Caplan syndrome), interstitial lung disease, pleuritis, pericarditis, anemia of chronic disease, neutropenia + splenomegaly (Felty syndrome: **S**ANITA—**S**plenomegaly, **A**nemia, **N**eutropenia, **T**hrombocytopenia, **A**rthritis [Rheumatoid]), AA amyloidosis, Sjögren syndrome, scleritis, carpal tunnel syndrome.



Gout**FINDINGS****A**

Acute inflammatory monoarthritis caused by precipitation of monosodium urate crystals in joints.

Risk factors: male sex, hypertension, obesity, diabetes, dyslipidemia, alcohol use. Strongest risk factor is hyperuricemia, which can be caused by:

- Underexcretion of uric acid (90% of patients)—largely idiopathic, potentiated by renal failure; can be exacerbated by certain medications (eg, thiazide diuretics).
- Overproduction of uric acid (10% of patients)—Lesch-Nyhan syndrome, PRPP excess, ↑ cell turnover (eg, tumor lysis syndrome).
- Combined mechanism—alcohol use and von Gierke disease.

✱

Crystals are needle shaped and ⊖ birefringent under polarized light (yellow under parallel light, blue under perpendicular light **A**). Serum uric acid levels may be normal during an acute attack.

SYMPTOMS**B**

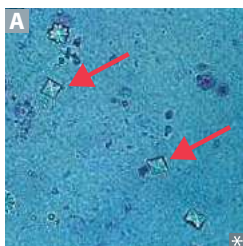
Asymmetric joint distribution. Joint is swollen, red, and painful. Classic manifestation is painful MTP joint of big toe (podagra). Tophus formation **B** (often on external ear, olecranon bursa, or Achilles tendon). Acute attack tends to occur after a large meal with foods rich in purines (eg, red meat, seafood), trauma, surgery, dehydration, diuresis, or alcohol consumption (↑ blood lactate from metabolism → ↑ resorption of uric acid → hyperuricemia).

TREATMENT

Acute: NSAIDs (eg, indomethacin), glucocorticoids, colchicine.

Chronic (preventive): allopurinol, probenecid.

For overproducers: urate lowering therapies such as xanthine oxidase inhibitors (eg, allopurinol, febuxostat).

Calcium pyrophosphate deposition disease

Formerly called pseudogout. Deposition of calcium pyrophosphate crystals within the joint space. Occurs in patients > 50 years old; both sexes affected equally. Usually idiopathic, sometimes associated with hemochromatosis, hyperparathyroidism, joint trauma.

Pain and swelling with acute inflammation (pseudogout) and/or chronic degeneration (pseudo-osteoarthritis). Most commonly affected joint is the knee.

Chondrocalcinosis (cartilage calcification) on x-ray.

Crystals are rhomboid and weakly ⊕ birefringent under polarized light (blue when parallel to light) **A**.

Acute treatment: NSAIDs, colchicine, glucocorticoids.

Prophylaxis: colchicine.

The **blue P's** of CPPD—**blue** (when **p**arallel), **p**ositive birefringence, calcium **p**rophosphate, **p**seudogout.

Systemic juvenile idiopathic arthritis

Systemic arthritis seen in < 16 years of age. Usually presents with daily spiking fevers, salmon-pink macular rash, arthritis (commonly 2+ joints). Associated with anterior uveitis. Frequently presents with leukocytosis, thrombocytosis, anemia, ↑ ESR, ↑ CRP.

Sjögren syndrome



Autoimmune disorder characterized by destruction of exocrine glands (especially lacrimal and salivary) by lymphocytic infiltrates. Predominantly affects females 40–60 years old.

Findings:

- Inflammatory joint pain
- Keratoconjunctivitis sicca (decreased tear production and subsequent corneal damage) → gritty or sandy feeling in eyes
- Xerostomia (↓ saliva production) → mucosal atrophy, fissuring of the tongue **A**
- Presence of antinuclear antibodies, rheumatoid factor (can be positive in the absence of rheumatoid arthritis), antiribonucleoprotein antibodies: SS-A (anti-Ro) and/or SS-B (anti-La)
- Dyspareunia
- Bilateral parotid enlargement

Anti-SSA and anti-SSB may also be seen in SLE.

A common 1° disorder or a 2° syndrome associated with other autoimmune disorders (eg, rheumatoid arthritis, SLE, systemic sclerosis).

Complications: dental caries; mucosa-associated lymphoid tissue (MALT) lymphoma (may present as parotid enlargement); ↑ risk of giving birth to baby with neonatal lupus.

Focal lymphocytic sialadenitis on labial salivary gland biopsy can confirm diagnosis.

Septic arthritis



S aureus, *Streptococcus*, and *Neisseria gonorrhoeae* are common causes. Usually monoarticular.

Affected joint is often swollen **A**, red, and painful. Synovial fluid purulent (WBC > 50,000/mm³).

Complications: osteomyelitis, chronic pain, irreversible joint damage, sepsis. Treatment: antibiotics, aspiration, and drainage (+/- debridement) to prevent irreversible joint damage.

Disseminated gonococcal infection—STI that presents as either purulent arthritis (eg, knee) or triad of polyarthralgia, tenosynovitis (eg, hand), dermatitis (eg, pustules).

Osteomyelitis

Chronic or acute infection of the bone; *S aureus* most common (overall), *S epidermidis* (prosthetics), *Salmonella* (sickle cell anemia), *P aeruginosa* (plantar puncture wounds). Spread is commonly hematogenous (usually in children, affecting the metaphysis of the long bones) or exogenous (usually in adults, post-traumatic, iatrogenic, or spread from nearby tissues). Pain, redness, swelling, fever, limping are common.

Diagnosis: x-ray (bone destruction and periosteal elevation if chronic), MRI, bone biopsy with cultures and blood cultures.

Treatment: antibiotics (+/- surgery).

**Seronegative
spondyloarthritis**

Arthritis without rheumatoid factor (no anti-IgG antibody). Strong association with HLA-B27 (MHC class I serotype). Subtypes (**PAIR**) share variable occurrence of inflammatory back pain (associated with morning stiffness, improves with exercise), peripheral arthritis, enthesitis (inflamed insertion sites of tendons, eg, Achilles), dactylitis (“sausage fingers”), uveitis.

Psoriatic arthritis

Associated with skin psoriasis and nail lesions. Asymmetric and patchy involvement **A**. Dactylitis and “pencil-in-cup” deformity of DIP on x-ray **B**.

Seen in fewer than 1/3 of patients with psoriasis.

**Ankylosing
spondylitis**

Symmetric involvement of spine and sacroiliac joints → ankylosis (joint fusion), uveitis, aortic regurgitation.

Bamboo spine (vertebral fusion) **C**. Costovertebral and costosternal ankylosis may cause restrictive lung disease. More common in males, with age of onset usually 20–40 years.

**Inflammatory bowel
disease**

Crohn disease and ulcerative colitis are often associated with spondyloarthritis.

Reactive arthritis

Classic triad:

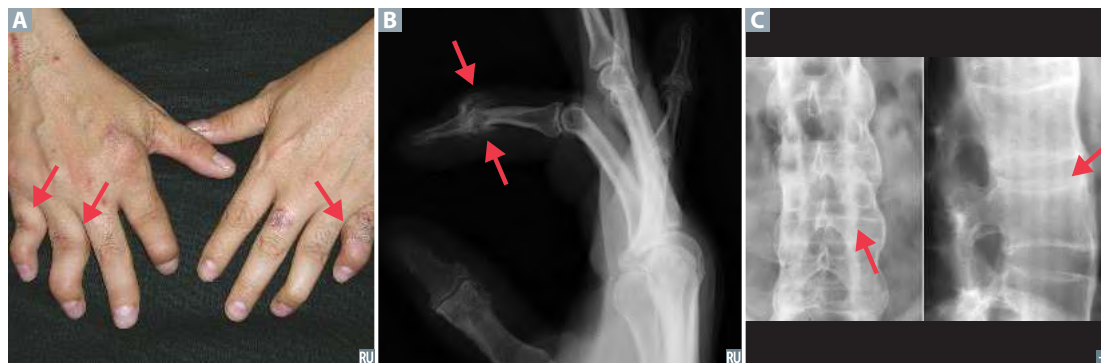
- **Conjunctivitis**
- **Urethritis**
- **Arthritis**

Commonly associated with hyperkeratotic skin lesions in the palms and soles (keratoderma blennorrhagica).

“Can’t **see**, can’t **pee**, can’t **bend my knee**.”

Associated with infections by *Shigella*, *Campylobacter*, *Salmonella*, *Chlamydia*, *Yersinia*.

“She **Caught Students Cheating Yesterday** and **overreacted**.”



Systemic lupus erythematosus

Systemic, remitting, and relapsing autoimmune disease. Organ damage primarily due to a type III hypersensitivity reaction and, to a lesser degree, a type II hypersensitivity reaction. Associated with deficiency of early complement proteins (eg, C1q, C4, C2) → ↓ clearance of immune complexes. Classic presentation: facial rash (spares nasolabial folds), joint pain, and fever in a female of reproductive age. ↑ prevalence in Black, Caribbean, Asian, and Hispanic populations in the US.



Libman-Sacks Endocarditis (LSE in SLE).

Lupus nephritis (glomerular deposition of DNA-anti-DNA immune complexes) can be nephritic or nephrotic (causing hematuria or proteinuria). Most common and severe type is diffuse proliferative.

Common causes of death in SLE: renal disease, infections, cardiovascular disease (accelerated CAD). Lupus patients die with redness in their cheeks.

In an anti-SSA ⊕ pregnant patient, ↑ risk of newborn developing **neonatal lupus** → congenital heart block, periorbital/diffuse rash, transaminitis, and cytopenias at birth.

RASH OR PAIN:

Rash (malar **A** or discoid **B**)

Arthritis (nonerosive)

Serositis (eg, pleuritis, pericarditis)

Hematologic disorders (eg, cytopenias)

Oral/nasopharyngeal ulcers (usually painless)

Renal disease

Photosensitivity

Antinuclear antibodies

Immunologic disorder (anti-dsDNA, anti-Sm, antiphospholipid)

Neurologic disorders (eg, seizures, psychosis)

Mixed connective tissue disease

Features of SLE, systemic sclerosis, and/or polymyositis. Associated with anti-U1 RNP antibodies (speckled ANA).

Antiphospholipid syndrome

1° or 2° autoimmune disorder (most commonly in SLE).

Diagnosed based on clinical criteria including history of thrombosis (arterial or venous) or recurrent abortion along with laboratory findings of lupus anticoagulant, anticardiolipin, anti-β₂ glycoprotein I antibodies.

Treatment: systemic anticoagulation.

Anticardiolipin antibodies can cause false-positive VDRL/RPR.

Lupus anticoagulant can cause prolonged PTT that is not corrected by the addition of normal platelet-free plasma.

Polymyalgia rheumatica

SYMPTOMS	Pain and stiffness in proximal muscles (eg, shoulders, hips), often with fever, malaise, weight loss. Does not cause muscular weakness. More common in females > 50 years old; associated with giant cell (temporal) arteritis.
FINDINGS	↑ ESR, ↑ CRP, normal CK.
TREATMENT	Rapid response to low-dose glucocorticoids.

Fibromyalgia

Most common in females 20–50 years old. Chronic, widespread musculoskeletal pain associated with “tender points,” stiffness, paresthesias, poor sleep, fatigue, cognitive disturbance (“fibro fog”). Normal inflammatory markers like ESR. Treatment: regular exercise, antidepressants (TCAs, SNRIs), neuropathic pain agents (eg, gabapentinoids).

Polymyositis/dermatomyositis

Nonspecific: ⊕ ANA, ↑ CK. Specific: ⊕ anti-Jo-1 (histidyl-tRNA synthetase), ⊕ anti-SRP (signal recognition particle), ⊕ anti-Mi-2 (helicase).

Polymyositis

Progressive symmetric proximal muscle weakness, characterized by endomysial inflammation with CD8+ T cells. Most often involves shoulders.

Dermatomyositis

Clinically similar to polymyositis, but also involves Gottron papules **A**, photodistributed facial erythema (eg, heliotrope [violaceous] edema of the eyelids **B**), “shawl and face” rash **C**, mechanic’s hands (thickening, cracking, irregular “dirty”-appearing marks due to hyperkeratosis of digital skin **D**). ↑ risk of occult malignancy. Perimysial inflammation and atrophy with CD4+ T cells.

**Myositis ossificans**

Heterotopic ossification involving skeletal muscle (eg, quadriceps). Associated with blunt muscle trauma. Presents as painful soft tissue mass. Imaging: eggshell calcification. Histology: metaplastic bone surrounding area of fibroblastic proliferation. Benign, but may be mistaken for sarcoma.

IgG4-related disease

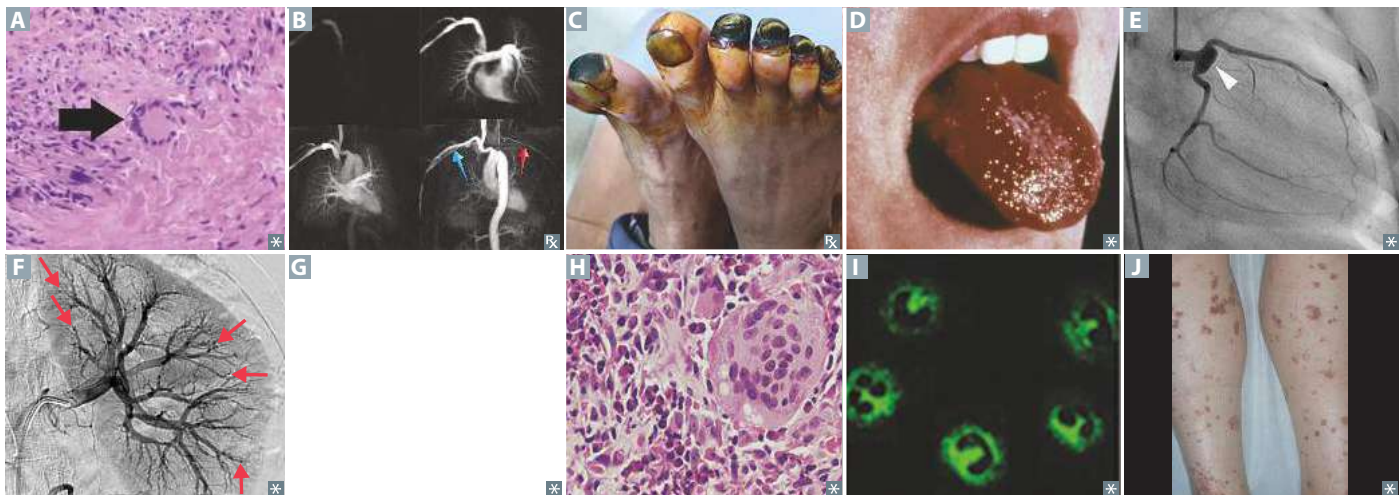
Immune-mediated spectrum of conditions, characterized by fibrosis and lymphoplasmacytic infiltrate, that can affect multiple organs. Patients usually have elevated serum IgG4 levels. Most common IgG4-related conditions are:

- Sialadenitis and dacryoadenitis
- Riedel thyroiditis
- Autoimmune pancreatitis
- Autoimmune aortitis (may lead to TAA, AAA)
- Retroperitoneal fibrosis (may affect the ureters and present with signs of acute kidney injury/CKD and/or hydronephrosis)

Vasculitides Inflammation and necrosis of blood vessels; either idiopathic or immune mediated (type III hypersensitivity).			
	EPIDEMIOLOGY	PRESENTATION/PATHOPHYSIOLOGY	NOTES
Large-vessel vasculitis			
Giant cell (temporal) arteritis	Females > 50 years old.	Unilateral headache, jaw claudication, temporal artery tenderness, blindness risk (due to anterior ischemic optic neuropathy). Granulomatous inflammation A ; affects temporal, vertebral, and ophthalmic arteries.	Associated with polymyalgia rheumatica. ↑↑ ESR/CRP; temporal artery biopsy. Treatment: high-dose glucocorticoids before biopsy.
Takayasu arteritis	Asian females < 40 years old.	“Pulseless disease” (weak upper extremity pulses), fever, night sweats, arthritis, myalgias. Granulomatous thickening and narrowing of aortic arch and proximal great vessels B .	↑ ESR. Treatment: glucocorticoids.
Medium-vessel vasculitis			
Buerger disease (thromboangiitis obliterans)	Heavy tobacco smoking history, males < 40 years old.	Leading to intermittent claudication, risk of gangrene C , Raynaud phenomenon, autoamputation of digits, or superficial nodular phlebitis.	Segmental thrombosing vasculitis with vein and nerve involvement. Treatment: smoking cessation.
Kawasaki disease	Asian children < 4 years old.	Bilateral nonexudative Conjunctivitis, Rash (polymorphous → desquamating), Adenopathy (cervical), Strawberry tongue (oral mucositis) D , Hand-foot changes (edema, erythema), fever (≥ 5 days).	CRASH and burn on a Kawasaki . Complications: coronary artery aneurysms E ; thrombosis or rupture can cause death. Treatment: IV immunoglobulin and aspirin.
Polyarteritis nodosa	Middle-aged males; 30% with hepatitis B seropositivity.	Fever, weight loss, abdominal pain, melena, hypertension, neurologic dysfunction, cutaneous eruptions, renal damage. Involves renal and visceral vessels, spares lungs; transmural inflammation with fibrinoid necrosis.	“String of pearls” appearance due to microaneurysms on arteriogram F . Treatment: glucocorticoids, cyclophosphamide. PAN usually affects the SKIN : S kin, K idneys, I ntestines (GI), N erves.
Small-vessel vasculitis (MPO-ANCA/p-ANCA)			
Microscopic polyangiitis	Typically affects middle-aged adults.	Necrotizing vasculitis involving lungs, kidneys, and skin. Pauci-immune glomerulonephritis (GN) G and palpable purpura.	MPO-ANCA/p-ANCA (anti-myeloperoxidase). Treatment: cyclophosphamide, glucocorticoids.
Eosinophilic granulomatosis with polyangiitis		Combination of asthma, eosinophilia, and systemic vasculitis. Eosinophilic infiltration → inflammation → peripheral neuropathy.	Formerly called Churg-Strauss syndrome. Granulomatous, necrotizing vasculitis with eosinophilia H . ↑ IgE level.

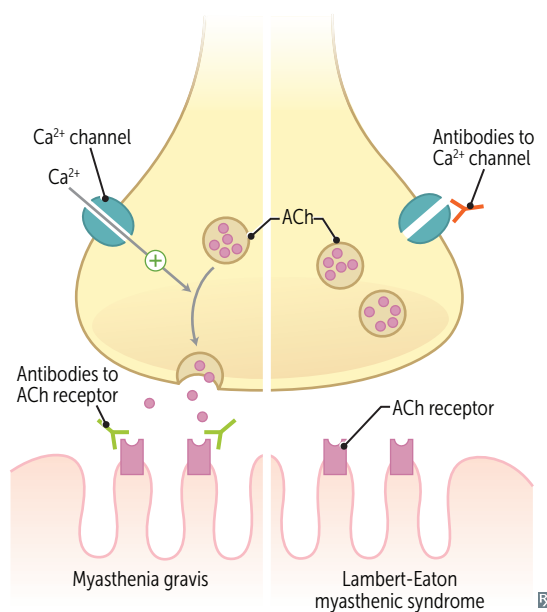
Vasculitides (continued)

	EPIDEMIOLOGY	PRESENTATION/PATHOPHYSIOLOGY	NOTES
Small-vessel vasculitis (c-ANCA)			
Granulomatosis with polyangiitis	Affects males and females equally, typically middle-aged adults.	Triad: lung, vessels, and renal involvement. Upper respiratory (nasal septum perforation, chronic sinusitis, otitis media, mastoiditis), lower respiratory (hemoptysis, dyspnea), renal (pauci-immune rapidly progressive GN).	c-ANCA I (PR3-ANCA). Treatment: glucocorticoids combined with rituximab, cyclophosphamide.
Small-vessel vasculitis (immune complex mediated)			
Hypocomplementemic urticarial vasculitis (anti-C1q vasculitis)	Often associated with SLE.	Presents as urticaria, purpuric rash, arthralgias, stomach pain, lung or ocular manifestations.	Labs show ↓ C1q complement and ↑ anti-C1q antibodies.
Mixed cryoglobulinemia	Often due to viral infections, especially HCV.	Triad of palpable purpura, weakness, arthralgias. May involve peripheral neuropathy and renal disease (eg, GN).	Cryoglobulins precipitate in the cold (mixed IgG and IgM immune complex deposition).
Immunoglobulin A vasculitis	Most common childhood vasculitis often follows URI.	Vasculitis 2° to IgA immune complex deposition. Classic triad: Hinge pain (arthralgias), stomach pain (abdominal pain associated with intussusception, palpable purpura on buttocks/legs J).	Formerly called Henoch-Schönlein purpura. Associated with IgA nephropathy (Berger disease). Treatment: supportive care, glucocorticoids.
Cutaneous small-vessel vasculitis		Palpable purpura, no visceral involvement. Immune complex-mediated leukocytoclastic vasculitis; late involvement indicates systemic.	Occurs 7–10 days after medication use (penicillins, cephalosporins, sulfonamides, phenytoin, allopurinol) or infections (eg, HCV, HIV).
All-vessel vasculitis			
Behçet syndrome	↑ incidence in Turkish, Eastern Mediterranean descent.	Recurrent oral and genital ulcers, uveitis, erythema nodosum. Triggered by HSV or parvovirus. Flares last 1–4 weeks.	Associated with HLA-B51.



Neuromuscular junction diseases

	Myasthenia gravis	Lambert-Eaton myasthenic syndrome
FREQUENCY	Most common NMJ disorder	Uncommon
PATHOPHYSIOLOGY	Autoantibodies to postsynaptic ACh receptor	Autoantibodies to presynaptic Ca^{2+} channel → ↓ ACh release; L comes before M
CLINICAL	Fatigable muscle weakness—ptosis; diplopia; proximal weakness; respiratory muscle involvement → dyspnea; bulbar muscle involvement → dysphagia, difficulty chewing Spared reflexes Worsens with muscle use	Proximal muscle weakness, autonomic symptoms (dry mouth, constipation, impotence) Hyporeflexia Improves with muscle use
ASSOCIATED WITH	Thymoma, thymic hyperplasia	Small cell lung cancer
ACHE INHIBITOR ADMINISTRATION	Reverses symptoms (pyridostigmine for treatment)	Minimal effect



Raynaud phenomenon

↓ blood flow to skin due to arteriolar (small vessel) vasospasm in response to cold or stress: color change from white (ischemia) to blue (hypoxia) to red (reperfusion). Most often in the fingers **A** and toes. Called **Raynaud disease** when 1° (idiopathic), **Raynaud syndrome** when 2° to a disease process such as mixed connective tissue disease, SLE, or CREST syndrome (limited form of systemic sclerosis). Digital ulceration (critical ischemia) seen in 2° Raynaud syndrome. Treat with calcium channel blockers.



Scleroderma

Systemic sclerosis. Triad of autoimmunity, noninflammatory vasculopathy, and collagen deposition with fibrosis. Commonly sclerosis of skin, manifesting as puffy, taut skin **A** without wrinkles, fingertip pitting **B**. Can involve other systems, eg, renal (scleroderma renal crisis; treat with ACE inhibitors), pulmonary (interstitial fibrosis, pulmonary HTN), GI (↓ peristalsis and LES tone → dysphagia, heartburn), cardiovascular. 75% female. 2 major types:

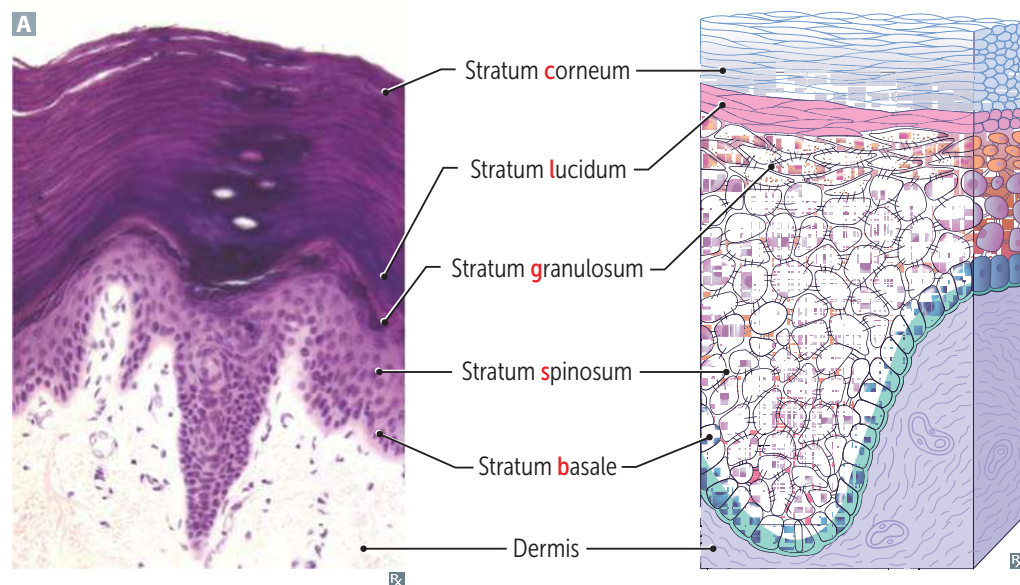
- **Diffuse scleroderma**—widespread skin involvement, rapid progression, early visceral involvement. Associated with anti-Scl-70 antibody (anti-DNA topoisomerase-I antibody) and anti-RNA polymerase III.
- **Limited scleroderma**—limited skin involvement confined to fingers and face. Also with **CREST** syndrome: **C**alcinosis cutis **C**, anti-**C**entromere antibody, **R**aynaud phenomenon, **E**sophageal dysmotility, **S**clerodactyly, and **T**elangiectasia. More benign clinical course.



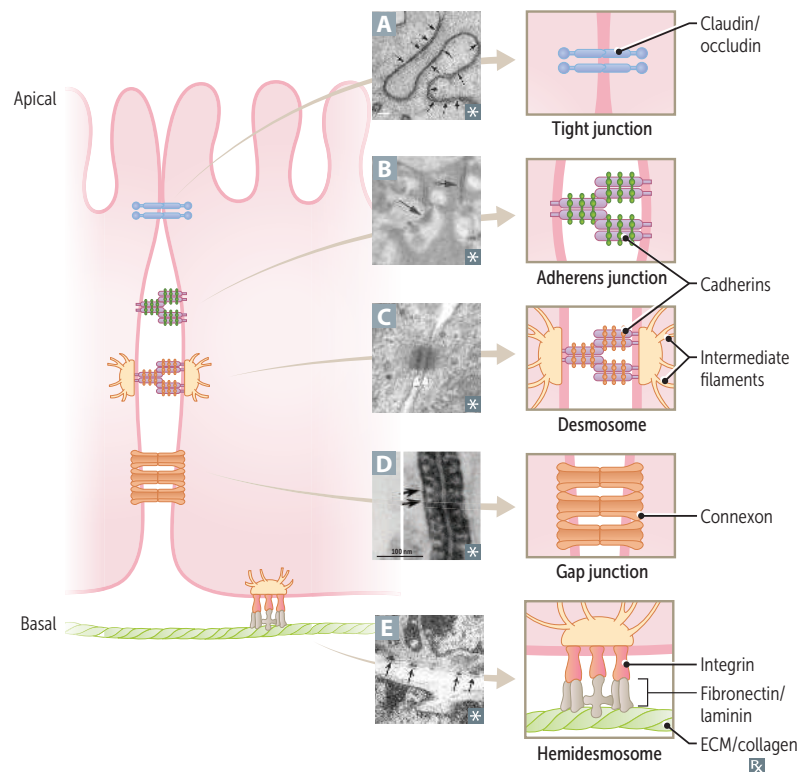
► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—DERMATOLOGY

Skin layers

Skin has 3 layers: epidermis, dermis, subcutaneous fat (hypodermis, subcutis).
Epidermal layers: **come, let's get sunburned**.



Epithelial cell junctions



Tight junctions (zonula occludens) **A**—prevents paracellular movement of solutes; composed of claudins and occludins.

Adherens junction (belt desmosome, zonula adherens) **B**—forms “belt” connecting actin cytoskeletons of adjacent cells with **cadherins** (Ca^{2+} -dependent **adhesion** proteins). Loss of E-cadherin promotes metastasis.

Desmosome (spot desmosome, macula adherens) **C**—structural support via intermediate filament interactions. Autoantibodies to desmoglein 3 +/- desmoglein 1 → pemphigus vulgaris.

Gap junction **D**—channel proteins called connexons permit electrical and chemical communication between cells.

Hemidesmosome **E**—connects keratin in basal cells to underlying basement membrane. Autoantibodies → **bullous** pemphigoid. (Hemidesmosomes are down “**bullo**.”)

Integrins—membrane proteins that maintain **integrity** of basolateral membrane by binding to collagen, laminin, and fibronectin in basement membrane.

Exocrine glands

Glands that produce substances other than hormones (vs endocrine glands, which secrete hormones) that are released through ducts to the exterior of the body. Can be merocrine (eg, salivary and sweat glands), apocrine (eg, mammary glands), or holocrine (eg, sebaceous glands).

